

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-22254326})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}1135 + \mathbf{Z}(408 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 22254326 = 0$, of discriminant -89017304 . Class group invariants $(170 \ 10 \ 5)$, class number 8500; the three stored generator ideals (HNF) are $\begin{pmatrix} 3731 & 33 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 105 & 68 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 274)y^3 + (-26s + 124004)y^2 + (-160s + 843464)y + (-7040s - 3184992)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 247460y^7 + 23768322y^6 + 1217122232y^5 \\ & + 38010888968y^4 + 705937346496y^3 + 8638102431040y^2 + 447616934 \\ & 28224y + 1113104177521664 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -1 \ 1 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right)s, \quad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25722962367664 = 2^4 \cdot 17 \cdot 281 \cdot 336547027$:

$$\begin{pmatrix} 6430740591916 & 152984799903 & 5784278660493 & 2156388354375 & 5388734372544 & 2249715465634 & 3281800058476 & 1874428435981 & 453143422599 & 3552468057593 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 384 factors with signed exponents of absolute value up to 342204713019417024708763852; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 274)y^3 + (-26s + 124004)y^2 + (-160s + 843464)y + (-7040s - 3184992)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 247460y^7 + 23768322y^6 + 1217122232y^5 \\ & + 38010888968y^4 + 705937346496y^3 + 8638102431040y^2 + 447616934 \\ & 28224y + 1113104177521664 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -1 \ 1 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944} \right) s, \quad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6102548079616 = 2^{10} \cdot 17 \cdot 761 \cdot 460657$:

$$\begin{pmatrix} 95352313744 & 21665950876 & 22620955972 & 2744162039 & 12983457041 & 40134592562 & 38973344485 & 11396481091 & 82133886882 & 61691623307 \\ 0 & 4 & 0 & 1 & 0 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 4 & 2 & 2 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 379 factors with signed exponents of absolute value up to 60044379493188783256086506; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 274)y^3 + (-26s + 124004)y^2 + (-160s + 843464)y + (-7040s - 3184992)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 549y^8 + 247460y^7 + 23768322y^6 + 1217122232y^5 + 38010888968y^4 + 705937346496y^3 + 8638102431040y^2 + 44761693428224y + 1113104177521664$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -1 \ 1 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right)s, \quad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $623352706772 = 2^2 \cdot 757 \cdot 205862849$:

$$\begin{pmatrix} 311676353386 & 217593510310 & 233217344991 & 89392999167 & 175814013287 & 155607433997 & 5837790543 & 124241934306 & 291997499085 & 267081267908 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 385 factors with signed exponents of absolute value up to 681695366085403329350887838; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1307)y^3 + (14s - 45179)y^2 + (-323s + 311148)y + (-4354s - 15506100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 + 2615y^8 - 92972y^7 + 24675229y^6 - 772853530y^5 + 21623637605y^4 - 76125109920y^3 + 1106615447646y^2 + 52945002665384y + 662321427559016$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ -1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right) s, \quad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $144573632899832 = 2^3 \cdot 211 \cdot 419 \cdot 1013 \cdot 201787$:

$$\begin{pmatrix} 36143408224958 & 9507047005235 & 7620525053315 & 21332342536837 & 435178638397 & 33744057194394 & 9902379060764 & 34843942065700 & 23616304529659 & 1087038676835 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 373 factors with signed exponents of absolute value up to 109536118334095704163433634; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = \begin{pmatrix} 1 & 0 & 2 \end{pmatrix}$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1307) y^3 + (14s - 45179) y^2 + (-323s + 311148) y + (-4354s - 15506100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2615y^8 - 92972y^7 + 24675229y^6 - 772853530y^5 + \\ & 21623637605y^4 - 76125109920y^3 + 1106615447646y^2 + 52945002665 \\ & 384y + 662321427559016 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ -1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right) s, \quad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1827774298048 = 2^6 \cdot 41 \cdot 293 \cdot 2377339$:

$$\begin{pmatrix} 57117946814 & 15180445462 & 31956016916 & 38387689546 & 15130530567 & 6607672904 & 15488499894 & 42971109092 & 7000575850 & 41391073478 \\ 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 373 factors with signed exponents of absolute value up to 96493234780934689694303003; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s + 1307)y^3 + (14s - 45179)y^2 + (-323s + 311148)y + (-4354s - 15506100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 2615y^8 - 92972y^7 + 24675229y^6 - 772853530y^5 + \\ & 21623637605y^4 - 76125109920y^3 + 1106615447646y^2 + 52945002665 \\ & 384y + 662321427559016 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ -1 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right)s, \quad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5451308556536 = 2^3 \cdot 17 \cdot 211 \cdot 521 \cdot 364621$:

$$\begin{pmatrix} 1362827139134 & 334216782147 & 483274228601 & 1124039577423 & 1136764613279 & 463585673214 & 1296261141754 & 1317876116578 & 111497716197 & 275895654537 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 368 factors with signed exponents of absolute value up to 67458245872218714261896352; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 3412y^3 + (-10s + 38166)y^2 + (-60s - 2693904)y + (26320s + 160387680)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 6823y^8 + 83156y^7 + 6177604y^6 + 65718384y^5 + 2 \\ & 1744501692y^4 - 1273411417248y^3 + 7865269518176y^2 - 93442608864 \\ & 3840y + 41140723099404800 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right)s, \quad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $49757027237108 = 2^2 \cdot 17 \cdot 731720988781$:

$$\begin{pmatrix} 24878513618554 & 2611879157078 & 4402196056015 & 15171208038109 & 10707226067999 & 22182452292499 & 15277990837132 & 10840657748104 & 17129997095411 & 13541848265407 \\ 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 306 factors with signed exponents of absolute value up to 2266523851800; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 3412y^3 + (-10s + 38166)y^2 + (-60s - 2693904)y + (26320s + 160387680)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 6823y^8 + 83156y^7 + 6177604y^6 + 65718384y^5 + 2 \\ & 1744501692y^4 - 1273411417248y^3 + 7865269518176y^2 - 93442608864 \\ & 3840y + 41140723099404800 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944} \right) s, \quad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55895963497600 = 2^7 \cdot 5^2 \cdot 17^3 \cdot 3555361$:

$$\begin{pmatrix} 6044113700 & 4827625660 & 5703618966 & 5116146198 & 4149037835 & 2328605976 & 4511401936 & 5605107662 & 4713555080 & 1182972825 \\ 0 & 34 & 0 & 27 & 20 & 0 & 27 & 11 & 8 & 8 \\ 0 & 0 & 34 & 22 & 1 & 2 & 26 & 23 & 25 & 9 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 299 factors with signed exponents of absolute value up to 2086164805237; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 3412y^3 + (-10s + 38166)y^2 + (-60s - 2693904)y + (26320s + 160387680)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 6823y^8 + 83156y^7 + 6177604y^6 + 65718384y^5 + 2 \\ & 1744501692y^4 - 1273411417248y^3 + 7865269518176y^2 - 93442608864 \\ & 3840y + 41140723099404800 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 1 \ 1 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right)s, \quad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10479774113920 = 2^7 \cdot 5 \cdot 113 \cdot 144908381$:

$$\begin{pmatrix} 327492941060 & 20730696176 & 213806419162 & 243180723120 & 290624740070 & 88676904391 & 1097384690 & 148328917835 & 272459835199 & 37327355968 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 305 factors with signed exponents of absolute value up to 3374984441326; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3448y^3 + 48sy^2 + 2250972y + 4752s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6896y^8 + 16390648y^6 + 35751264192y^4 + 15219120431376y^2 + 502536151586304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right)s, \quad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65730985358445 = 3^2 \cdot 5 \cdot 11 \cdot 73 \cdot 1819039307$:

$$\begin{pmatrix} 21910328452815 & 21443205004476 & 18305065303343 & 15108083457521 & 9087095186500 & 4367738330771 & 3448605092181 & 3966745825383 & 2991267132982 & 14858222736136 \\ 0 & 3 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 118 factors with signed exponents of absolute value up to 10049343475; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3448y^3 + 48sy^2 + 2250972y + 4752s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6896y^8 + 16390648y^6 + 35751264192y^4 + 15219120431376y^2 + 502536151586304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right)s, \quad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $201757417959375 = 3^2 \cdot 5^5 \cdot 17 \cdot 241 \cdot 421 \cdot 4159$:

$$\begin{pmatrix} 1614059343675 & 1482987640910 & 710592456315 & 1583357826880 & 1025715615623 & 381656838563 & 238205790518 & 1063541738071 & 571177889553 & 815939166601 \\ 0 & 5 & 0 & 0 & 0 & 2 & 2 & 2 & 0 & 4 \\ 0 & 0 & 5 & 0 & 2 & 0 & 1 & 4 & 4 & 0 \\ 0 & 0 & 0 & 5 & 2 & 3 & 4 & 2 & 4 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 11911056261; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 4 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3448y^3 + 48sy^2 + 2250972y + 4752s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6896y^8 + 16390648y^6 + 35751264192y^4 + 15219120431376y^2 + 502536151586304$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right)s, \quad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1683124114212 = 2^2 \cdot 3^9 \cdot 17 \cdot 67 \cdot 137^2$:

$$\begin{pmatrix} 8426322 & 3218541 & 6901113 & 6821134 & 2201875 & 5259342 & 760 & 3046288 & 3048563 & 5851911 \\ 0 & 411 & 354 & 274 & 132 & 149 & 272 & 71 & 232 & 23 \\ 0 & 0 & 9 & 3 & 8 & 8 & 3 & 5 & 6 & 8 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 11758747692; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 2831)y^3 + (-7s + 216964)y^2 + (250s + 2567748)y + (344s + 21798120)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 8y^9 + 5678y^8 + 411280y^7 + 33668671y^6 + 1563064988y^5 + 51400880486y^4 + 1144437592296y^3 + 17335862947848y^2 + 115771902139520y + 477791523455936$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right) s, \quad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $715380577792 = 2^9 \cdot 31^3 \cdot 46901$:

$$\begin{pmatrix} 5815724 & 3898002 & 4407890 & 1678234 & 3070360 & 900940 & 1139524 & 3303839 & 5012874 & 5044946 \\ 0 & 62 & 0 & 28 & 40 & 34 & 36 & 43 & 20 & 26 \\ 0 & 0 & 62 & 48 & 36 & 8 & 6 & 45 & 18 & 39 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 316 factors with signed exponents of absolute value up to 11518664377183; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 2831)y^3 + (-7s + 216964)y^2 + (250s + 2567748)y + (344s + 21798120)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 5678y^8 + 411280y^7 + 33668671y^6 + 1563064988y^5 \\ & + 51400880486y^4 + 1144437592296y^3 + 17335862947848y^2 + 115771 \\ & 902139520y + 477791523455936 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right) s, \quad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $151644131083504 = 2^4 \cdot 17 \cdot 31 \cdot 311 \cdot 57827527$:

$$\begin{pmatrix} 37911032770876 & 35495499484903 & 13356946055389 & 19446838963646 & 17133079054323 & 6032181862718 & 26102857327172 & 7322424794045 & 440213513907 & 28451441224843 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 316 factors with signed exponents of absolute value up to 12920316257821; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 2831)y^3 + (-7s + 216964)y^2 + (250s + 2567748)y + (344s + 21798120)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 5678y^8 + 411280y^7 + 33668671y^6 + 1563064988y^5 \\ & + 51400880486y^4 + 1144437592296y^3 + 17335862947848y^2 + 115771 \\ & 902139520y + 477791523455936 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right)s, \quad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $14072221709552 = 2^4 \cdot 31^2 \cdot 167 \cdot 2341^2$:

$$\begin{pmatrix} 24238714 & 13861061 & 18923111 & 17119795 & 23318803 & 22599051 & 9729395 & 17491024 & 4606488 & 2240226 \\ 0 & 72571 & 26282 & 59587 & 11705 & 43573 & 43360 & 16012 & 46367 & 11926 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 317 factors with signed exponents of absolute value up to 15969045174199; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 1 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 176y^3 + 18sy^2 - 629825y - 1026s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 352y^8 - 1228674y^6 + 7432100024y^4 - 425306254511y^2 + 23426594876376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right)s, \quad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4042709872992 = 2^5 \cdot 3^5 \cdot 7 \cdot 313 \cdot 237287$:

$$\begin{pmatrix} 9358124706 & 7253214804 & 758904330 & 1488470346 & 4488292088 & 2993093001 & 8388389922 & 8413685229 & 3439242581 & 4416695629 \\ 0 & 6 & 0 & 0 & 2 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 6 & 0 & 4 & 0 & 2 & 3 & 2 & 1 \\ 0 & 0 & 0 & 6 & 4 & 4 & 3 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 154 factors with signed exponents of absolute value up to 292647140; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 176y^3 + 18sy^2 - 629825y - 1026s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 352y^8 - 1228674y^6 + 7432100024y^4 - 425306254511y^2 + 23426594876376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944} \right) s, \quad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $132782106393216 = 2^7 \cdot 3^2 \cdot 7^4 \cdot 29 \cdot 1655377$:

$$\begin{pmatrix} 395184840456 & 214112868564 & 2290539928 & 125438298108 & 283117162830 & 373445348788 & 77019394157 & 241016247860 & 91004307413 & 120539806415 \\ 0 & 21 & 14 & 15 & 8 & 18 & 7 & 18 & 10 & 3 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 8 & 7 & 4 & 7 & 4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 154 factors with signed exponents of absolute value up to 3011982; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 176y^3 + 18sy^2 - 629825y - 1026s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 352y^8 - 1228674y^6 + 7432100024y^4 - 425306254511y^2 + 23426594876376$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -1 \ -1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right)s, \quad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $23853906432 = 2^9 \cdot 3^{11} \cdot 263$:

$$\begin{pmatrix} 511272 & 59862 & 235812 & 414250 & 65256 & 316837 & 142624 & 164558 & 74929 & 323076 \\ 0 & 6 & 0 & 2 & 0 & 2 & 4 & 4 & 5 & 0 \\ 0 & 0 & 12 & 6 & 6 & 10 & 2 & 2 & 6 & 6 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 54 & 9 & 46 & 46 & 18 & 46 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 72831596; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.